

Sunwoo Hong

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RESEARCH INTERESTS

I work on **closing the gap between what people do easily and what models still cannot**: people reason instead of matching patterns, learn from very little, and act in a physical world they understand. I pursue this through **reasoning**, **efficient learning**, and **world models for embodied agents**, by finding the assumption that holds a model class back and replacing it.

Keywords: reasoning, world models, RL, diffusion language models, embodied AI, data-efficient learning

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) M.S. in Artificial Intelligence, Graduate School of AI Advisor: Prof. Se-Young Yun	Seoul, South Korea <i>Feb 2027 (expected)</i>
Seoul National University (SNU) B.S. in Electrical and Computer Engineering	Seoul, South Korea <i>Feb 2024</i>
Sejong Academy of Science and Arts Inaugural Student Council President; Government-designated academy for gifted students	Sejong, South Korea <i>Feb 2018</i>

PUBLICATIONS

- *: equal contribution (random order), †: corresponding author, C: conference, W: workshop, U: under review, P: in preparation, D: domestic
- [C2] S. H. Kim*, **Sunwoo Hong***, Y. Choi, C.-H. Chao, S.-Y. Yun†, and R. G. Krishnan†. Pivot-SD: Efficient Self-Distillation for Masked Diffusion Language Models. **Empirical Methods in Natural Language Processing (EMNLP), Main Conference**, 2026. **Oral presentation**. [\[project\]](#)
 - [C1] S. H. Kim*, **Sunwoo Hong***, H. Jung, Y. Park, and S.-Y. Yun. KLASS: KL-Guided Fast Inference in Masked Diffusion Models. **Neural Information Processing Systems (NeurIPS)**, 2025. **Spotlight (top 3.2% of submissions)**. [\[pdf\]](#) [\[code\]](#)
 - [U1] **Sunwoo Hong***, S. H. Kim*, C.-H. Chao, Y. Choi, S.-Y. Yun, and R. G. Krishnan. Logical Equivariance in Diffusion Language Models. **Under review**. Earlier version presented at **ICML Workshop** on Foundations of Deep Generative Models, 2026. [\[pdf\]](#)
 - [W1] C. Park, **Sunwoo Hong**, K. Kim, Y. Park, A. W. Olson, and A. Pesaraghader†. Route-Guided CBF-QP Repair for Safe Vision-Language-Action Models. **RSS Workshop** on Trustworthy Embodied Foundation Models, 2026. [\[pdf\]](#) [\[project\]](#) [\[code\]](#)
 - [P1] **Sunwoo Hong**. Do World Models Need Clocks? In preparation for submission to CVPR 2027.
 - [D1] **Sunwoo Hong**, Y. Kim, S. Jung, J. Oh, and S.-Y. Yun. Memory Reduction via Ternary Quantization in Reinforcement Learning. Korea Institute of Military Science and Technology (KIMST) Conference, 2025.

RESEARCH EXPERIENCE

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|---|---------------------------|
| Graduate Research Assistant, OSI Lab, KAIST AI
Advisor: Prof. Se-Young Yun [page] | <i>Feb 2025 – Present</i> |
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- Building a world model that steps from one physical event to the next (impact, release, rest), predicting when it happens and the state after it, so a planner compares actions over a few events, not a long rollout of frames [P1].
 - Found that the KL divergence between consecutive denoising steps tells a masked diffusion model which tokens are already settled. Unmasking those in parallel, with no training, gives up to $2.78\times$ faster generation while improving on greedy decoding [C1].

Visiting Researcher, University of Toronto & Vector Institute*Jan 2026 – Jul 2026**Advisor: Prof. Rahul G. Krishnan [page]*

- Showed that diffusion language models solve about 99% of Sudoku states in at least one of eight rotated or reflected views but only 13–24% in all eight, and proved that hard-wiring this symmetry into attention leaves it almost no capacity, so consistency must be learned [U1].
- Found that a few early commitments decide most of a diffusion language model’s output. Training on those alone (10 tokens per trajectory, under 4%) beats full-sequence fine-tuning and RL runs with ten times the data, at one-sixth to one-eighth of the wall-clock time [C2].

Research Collaborator, LG Electronics Toronto AI Lab*Jan 2026 – Jul 2026**Mentor: Dr. Ali Pesaranghader [page]*

- Safety filters keep a robot policy from colliding but push it off its task path. Co-developed a module that routes it around the obstacle and back, improving collision avoidance and task success over AEGIS on SafeLIBERO [W1].

Graduate Research Assistant, MLML Lab, KAIST School of Computing*Feb 2024 – Jan 2025**Advisor: Prof. Sungjin Ahn [page]*

- Designed a compact 1D tokenizer that compresses video into a short sequence of global tokens instead of a spatial grid, to model scene-level dynamics in video world models.
- Studied why block-slot object-centric models fail to separate factors such as shape and color, and built out-of-distribution tests that measure compositional generalization.

Undergraduate Researcher, AISys Lab, SNU*Jan 2023 – May 2023**Advisor: Prof. Jinho Lee (B.S. thesis) [page]*

- Built pipeline-parallel training schedules for knowledge distillation that remove idle time between teacher and student passes: up to $1.45\times$ faster training or $1.55\times$ better memory efficiency than a GPipe baseline.

Research Intern, Human-Centered Computer Systems Lab, SNU*Jun 2022 – Sep 2022**Advisor: Prof. Youngki Lee [page]*

- Built a server-driven adaptive video streaming pipeline for deep learning inference, reducing bandwidth by up to 70% while maintaining detection accuracy.

Undergraduate Research Program (URP), SNU*May 2021 – Dec 2021**Advisor: Prof. Kyomin Jung [page]*

- Developed a GAN-based model that transfers a sign-language interpreter’s motion, extracted as skeletons, onto the speaker in a video, so that the speaker appears to sign.

AWARDS AND HONORS

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| • Oral Presentation, EMNLP 2026 | <i>Oct 2026</i> |
| • Excellence Selection, GARL 2026 Challenge (National AI Research Lab, Korea) [U1] | <i>Jul 2026</i> |
| • Spotlight (top 3.2% of submissions), NeurIPS 2025 | <i>Dec 2025</i> |
| • Full Scholarship, Choi Hyuk-Young Scholarship Foundation | <i>Mar 2019 – Present</i> |
| • Minister of Education Award (1st place), national STEAM R&E research competition | <i>Nov 2015</i> |

TEACHING EXPERIENCE

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| • Teaching Assistant, Geometric Deep Learning (AI810), KAIST | <i>Spring 2025</i> |
| • Teaching Assistant, Reinforcement Learning (CS672), KAIST | <i>Fall 2024</i> |
| • Teaching Assistant, Freshman Seminar: Writing (E51.101), Seoul National University | <i>Fall 2021</i> |

ACADEMIC SERVICES**Reviewer** ACL Rolling Review (August 2026), ICLR 2027**MILITARY SERVICE**

- Republic of Korea Air Force, Sergeant (honorable discharge); information systems specialist *Aug 2019 – May 2021*